INCH-POUND

A-A-52298
February 20, 1992

COMMERCIAL ITEM DESCRIPTION

TRIPOD, SURVEYING: EXTENSION-LEG-TYPE

The General Services Administration has authorized the use of this commercial item description in preference to Military specification MIL-T-14091E.

This description covers an extension-leg-type tripod for surveying instruments.

Salient characteristics.

Design. The extension-leg-type surveying tripod is designed for use with theodolites, levels, electronic distance measuring devices, and other surveying instruments. The tripod shall be the extension-leg-type and shall be equipped with a threaded instrument fastener, triangular shape head, a headcap, a carrying sling, and a case attached to the tripod containing a wrench, plumb bob, cord, cord adjuster and cord centering plumb holding device. When a bayonet-type plumbline holder is provided, an additional standard plumbline holder shall be furnished for arctic use (see figure 1).

Classification. The classification of extension-leg-type surveying tripods shall be as follows:

Type - A - with telescopic plumbing rod.
Type - B - without telescopic plumbing rod.

Material:
Material shall be as described herein.

Metal parts. All metal components of the tripod shall be nonmagnetic, except for the corrosion resistant steel (CRSE) points.

Webbing. Cotton or synthetic material webbing shall be medium weight with self texture, dyed, and treated to be water repellent and mildew resistant (Copper 8-quinolinolate).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and other data which may improve this document should be sent by letter to: Commander, U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 6675

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Wood. One of the following species of hardwood shall be used: White ash, European beech, yellow birch, pecan hickory, or hard maple. Slope of grain shall not exceed 1-inch in 15 inches. Wood shall be free from defects such as knots, checks.

Tripod head. The tripod head as shown in figure 1 shall be approximately triangular shaped; each side shall be 6.75 inches long ±0.75-inch. A 2.25 to 2.50-inch diameter hole shall be provided in the tripod head; the center of the hole shall be located at the bisectors of the angles formed by the sides of the tripod head. Bearing blocks shall be provided at each vertex for hinge, pin-type bearings, not less than 0.375-inch long and 0.375-inch in diameter. A single stud bolt and a fiber washer shall be located under the tripod head at each vertex to apply pressure on the bearing blocks for the adjustment of tension on the hinges.

Instrument fastener. An instrument fastener as show in figure 1 shall hold the surveying instrument in place on top of the tripod head. The fastener shall be attached to the tripod head by a sliding arrangement. The sliding arrangement shall be pivoted and fastened at one end to the bottom of the tripod head (milled head); allowing the instrument to be positioned anywhere within the perimeter of the hole in the center of the tripod head. Threads on the fastener shall be 0.625-11 UNC-2A conforming to FED-STD-H28 and FED-STD-H28/2. When the instrument fastener is pushe upward to its extremity, it shall protrude above the tripod head not les. than 0.375-inch and not more than 0.50-inch. When not in use, the instrument fastener shall drop below the surface of the head. Provision shall be made for suspending a plumb bob from the instrument fastener center in such a way that the pivot point of the plumb bob line itself or of the plumb bob line extended by the plumbline device shall be within 0.50-inch, ±0.06-inch, of the top of the tripod head when fastened to an instrument. The total unobstructed plumbline swing shall be not less than ±5 degrees from the vertical. The plumbline holding device shall be a quick-disconnect type so that the plumb bob can be removed for use of the optical plumb. The grip portion of the instrument fastener shall be long enough to permit the fastening of an instrument by the operator wearing arctic gloves. The grip portion of the plumbline holding device shall be long enough to permit the insertion of the plumbline holding device into the instrument fastener by an operator whether he is or is not wearing arctic gloves.

Hinges. The hinges shall provide a firm connection between the legs and the tripod head. Pin-type bearings of the hinge not less than 0.375-inch long and not less than 0.375-inch in diameter shall connect the hinge to the bearing blocks of the tripod head.

Legs. The three tripod legs shall be the extension-leg type, enabling the lower part of the leg to slide into the upper part of the leg. When the leg is extended to its full length, the lower part of the leg shall lap not less than 2.5 inches into the upper part of the leg. The extended

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represent clearly the operation and the function of the electrical

length, measured from the center of the hinge to tip of the steel point shall be 70 inches, ±2 inches. The closed length of the leg, measured from the center of the hinge to the tip of the steel point, shall be 44 inches, ±2 inches. A cotton webbing or leather strap with buckle shall be attached to the outside of one leg to hold the three legs together.

Upper part. The upper part of the tripod leg shall consist of two wooden dowels or two aluminum dowels, a metal spacer, or other means to hold the battery box, and a metal band with a clamping device for holding the lower part of the leg in place when adjusting the tripod to the required height.

Lower part. The lower part of the tripod leg shall consist of two wooden dowels or two aluminum dowels, or a single oblong section made from aluminum or wood, two metal stops, a metal foot spur, and a metal shoe with a steel point.

Metal shoe. The metal shoe shall be 4 to 7 inches long and a thickness and width to form a socket-type arrangement to accommodate the lower section. The lower end shall be seated in the socket of the metal shoe. The lower end of the metal shoe shall be tapered.

Steel point. The top cross section of the steel point shall match the bottom cross section of the metal shoe as to dimensions and configuration. The point shall be fastened to the metal shoe except when a replaceable point is supplied, the top of the point may have a hex-shaped cross section of proper dimensions so that the wrench supplied as shown in figure 1 can be used to tighten or remove the replaceable points. There shall be a taper beginning with the taper on the metal shoe to the tip of the point. The length of the entire taper, including metal shoe and the point shall be 3 to 8 inches. The point shall be CRES and hardened to RC 40-60.

Foot spur. There shall be a foot spur on the outside of the metal shoe on each leg and it shall extend out from the face of the shoe 1.50 inches ±0.05-inch.

Construction. When wood is used in the construction of the legs, all joints shall be secured with oval-head wood screws countersunk into the metal part of the joint. At joints that require tightening, hex head lag screws or self-locking hex head bolts and nuts may be used. When aluminum dowels are used in the construction of the legs, the joints shall be secured with aluminum rivets, peened to a button head. All joints shall be secured with waterproof cement to prevent penetration of moisture.

Headcap. The headcap shall be nonmagnetic metal or high-impact plastic. Provision shall be made for fastening the headcap to one of the metal

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spacers and for holding the headcap flat against the tripod leg. The headcap shall have provisions to be attached to cover the tripod head with the instrument fastener. If high impact plastic is used as the headcap material, a nonmagnetic metal thread insert shall be molded in the headcap to mate with the instrument fastener.

Battery box attachment. One of the metal spacers attached to the upper dowels of the tripod leg shall be used as a bracket for holding the battery box to the outside of the tripod leg. The bracket shall be offset 0.375 inch, ±0.006 inch, from the tripod leg. The upper edge of the bracket shall be notched, 1.375 inches long by 0.125 inch deep, ±0.062 inch.

Plumb bob. The plumb bob shall be constructed of brass, with replaceable points. The plumb bob shall be furnished with not less than 7.50 feet of braided linen cord, cord adjuster and plumbline holding device; all assembled for operation.

Wrench. A wrench shall be furnished to permit the adjustment of all bolts and nuts.

Case. A case made from leather, cotton webbing, or high-impact plastic shall hold the wrench and plumb bob assembly. The case shall be attach to the inside of one of the tripod legs. The case shall not hinder the full movement of any part of the tripod leg.

Carrying sling. An adjustable, padded carrying sling of cotton webbing shall be fastened to one leg of the tripod. The sling shall be sufficient length for carrying the tripod on the shoulder.

Telescopic plumbing rod. When specified, a telescopic plumbing rod with a mounted circular level shall be furnished. The plumbing rod shall be nonmagnetic metal. The level shall have a sensitivity of not less than 20 minutes per 2 millimeters. The rod shall be graduated so that the height of the surveying instrument may be determined to the nearest 0.1 inch. The rod shall be attachable to the instrument fastener by a quick-release lock. Provision shall be made for firm attachment of the rod to one leg of the tripod. The rod shall be fabricated so that the inner and outer members shall move over the fully extended length without binding or bending.

Weight. The weight of the tripod excluding the plumb bob, case, wrench, and carrying sling, shall not exceed 15.5 pounds.

Identification marking The tripod shall be identified with the manufacturer's identification.

Treatment and painting. The portions of the tripod normally painted internally or externally shall be cleaned, treated and painted. The

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instrument contact surface of the tripod head and the threaded portion of the instrument fastener shall not be painted.

Workmanship. All edges shall be rounded or beveled. All joints shall be firm and dowels shall be firmly seated in their sockets. Dowels shall be straight. Metal and wood shall be free from cracks, scratches, or splits. Screws and boltholes shall be drilled and shall have burrs removed. Bolts, nuts, screws, and rivets shall be tight. There shall be no binding when shortening or lengthening the tripod legs. The tripod shall not wobble and legs shall not bend when the tripod is extended to its full length. Paint shall be evenly applied on all painted surfaces.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Quality assurance: The tripod shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Material not as specified.
2. Metal parts missing or not as specified.
3. Webbing not as specified.
4. Wood type not as specified.
5. Tripod head not as specified.
6. Instrument fastener not as specified.
7. Plumbline holding device quick-connect/disconnect not as specified.
8. Screw threads not as specified.
9. Hinges not as specified.
10. Tripod legs not as specified.
11. Metal shoe not as specified.
12. Steel point not as specified.

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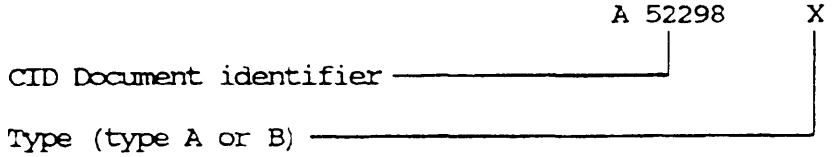
13. Foot spur not as specified.
14. Construction not as specified.
15. Headcap not as specified.
16. Battery box attachment not as specified.
17. Plumb bob assembly not as specified.
18. Wrench not as specified.
19. Case not as specified.
20. Carrying sling not as specified.
21. Telescopic plumbing rod not as specified.
22. Weight not as specified.
23. Identification marking missing, incomplete, or illegible.
24. Treatment and painting not as specified.
25. Workmanship not as specified.
26. Packaging and marking not as specified.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the contractor's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification numbers. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Part or identifying number (PIN). The part or identifying number for tripod shall be as follows:



Notes: The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. The ordering activity shall specify type required.
3. When the commercial item description is used for acquisition, the commercial item certification clause must appear in the solicitation.

FED-STD-376, FED-STD-H28, and FED-STD-H28/2 are available from the

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Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

The following are suggested sources of supply for this item description (Competition is not limited to these sources.):

1. Leica, Inc, 111 Deer Lake Rd., Deerfield, IL 60015.
2. Seiler Instrument and Manufacturing Co., Inc., 170 E. Kirkham Ave., St. Louis, MO 63119.
3. John Woods Survey Equipment Limited, 2380 Midland Ave., Scarborough, Ontario, Canada, MIS 1P8.
4. The Lietz Co, 9111 Barton St., P. O. Box 2934, Overland Park, KA 66201.

The preparing activity for this item description is the US Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

MILITARY INTEREST:

Custodians
Army - ME

Review activity
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CIVIL AGENCY COORDINATING ACTIVITIES:
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PREPARING ACTIVITY:
Army - ME

Project 6675-0899

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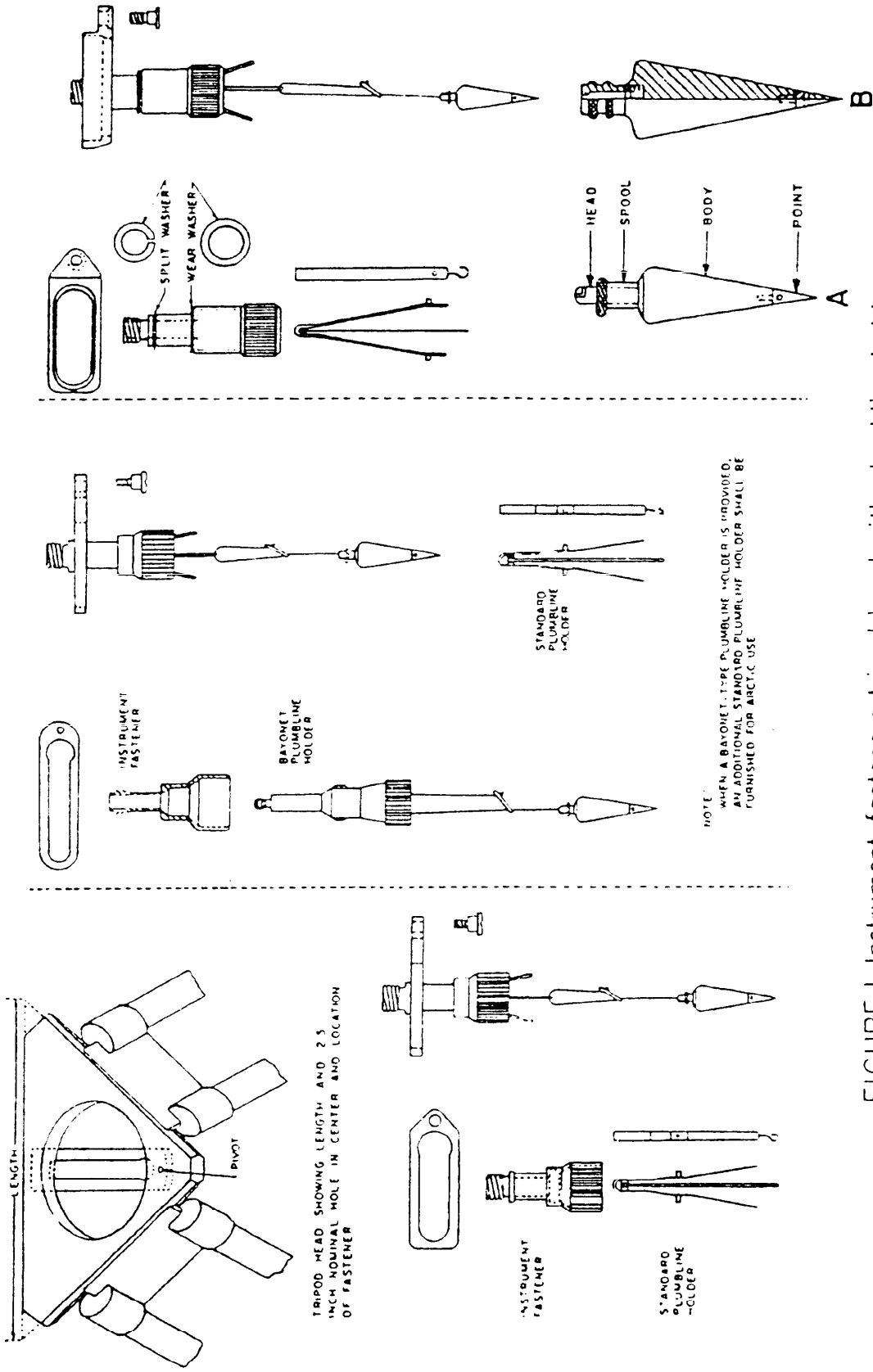


FIGURE 1. Instrument fasteners tripod head, with plumbline holders.

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